

N 705

Seat No.

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2025 VII 03 1100 – **N 705** – SCIENCE AND TECHNOLOGY (72) – PART I (E)
(REVISED COURSE)

Time : 2 Hours

(Pages 8)

Max. Marks : 40

Note :— (i) **All questions are compulsory.**

(ii) **Use of a calculator is not allowed.**

(iii) **The numbers to the right of the questions indicate full marks.**

(iv) **In case of MCQs [Q. No. 1(A)] only the first attempt will be evaluated and will be given credit.**

(v) **Scientifically correct, labelled diagrams should be drawn wherever necessary.**

(vi) **Each new question should be started on the new page.**

1. (A) **Choose the correct alternative :** **5**

(i) Alkaline earth metals have valency 2. This means that their position in the modern periodic table is in

(A) Group 2

(B) Group 16

(C) Period 2

(D) *d*-block

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- (ii) When the positive charge on an atom or an ion increases or the negative charge on them decreases it is called
- (A) Reduction
 - (B) Oxidation
 - (C) Displacement
 - (D) Decomposition
- (iii) device is used to produce electricity.
- (A) Electric motor
 - (B) Galvanometer
 - (C) Electric generator
 - (D) Voltmeter
- (iv) Light changes its direction when going from one transparent medium to another transparent medium. This is called the of light.
- (A) Reflection
 - (B) Dispersion
 - (C) Refraction
 - (D) Absorption

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(v) Height of the High earth orbit from the surface of the earth is

(A) = 20,000 km

(B) = 27,000 km

(C) < 30,780 km

(D) > 35,780 km

(B) Answer the following questions :

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(i) Find the odd one out :

Fuse wire, bad conductor, rubber gloves, generator.

(ii) Complete the co-relation.

Red light wavelength close to : 700 nm :: Violet light wavelength close to :

(iii) Match the pair :

Group 'A'

Farsightedness

Group 'B'

- (a) Faraway object can be seen clearly
- (b) Nearby object can be seen clearly
- (c) Faraway object cannot be seen clearly

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(iv) Write the electronic configuration of 'Aluminium'.

(v) State true or false :

The escape velocity on the moon is less than that on the earth.

2. (A) **Give scientific reasons (any *two*) :** **4**

(i) Sodium is always kept in kerosene.

(ii) We cannot clearly see an object kept at a distance less than 25 cm from the eye.

(iii) Tungsten metal is used to make a solenoid type coil in an electric bulb.

(B) **Answer the following questions (any *three*) :** **6**

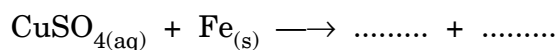
(i) An object takes 5 seconds to reach the ground from a height of 5 m on a planet. What is the value of 'g' on the planet ?

(ii) An element has its electronic configuration as 2, 8, 2. Now answer the following questions :

(a) What is the atomic number of this element ?

(b) To which period does this element belong ?

(iii) Complete the following reaction and give name of products :



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(iv) Define specific heat of an object. Write its SI and CGS units.

(v) Draw the electron-dot structure of methane (CH_4) and Ethane (C_2H_6).

3. Answer the following questions (any *five*) :

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(i) Explain the concept 'escape velocity' in detail.

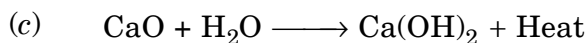
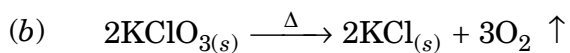
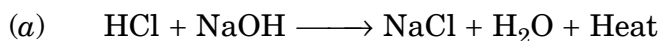
(ii) Answer the following sub-questions regarding the modern periodic table :

(a) Write modern periodic law.

(b) Write the number of periods and groups.

(c) What is the valency of halogens ?

(iii) State whether the following reactions are endothermic or exothermic :



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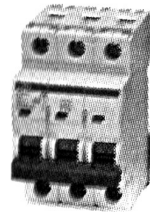
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(iv) Identify figure and write *one* use of it :

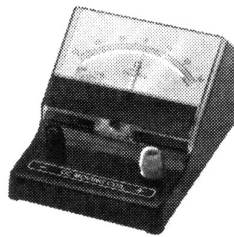
(a)



(b)



(c)



(v) How much heat energy is necessary to raise the temperature of 5 kg

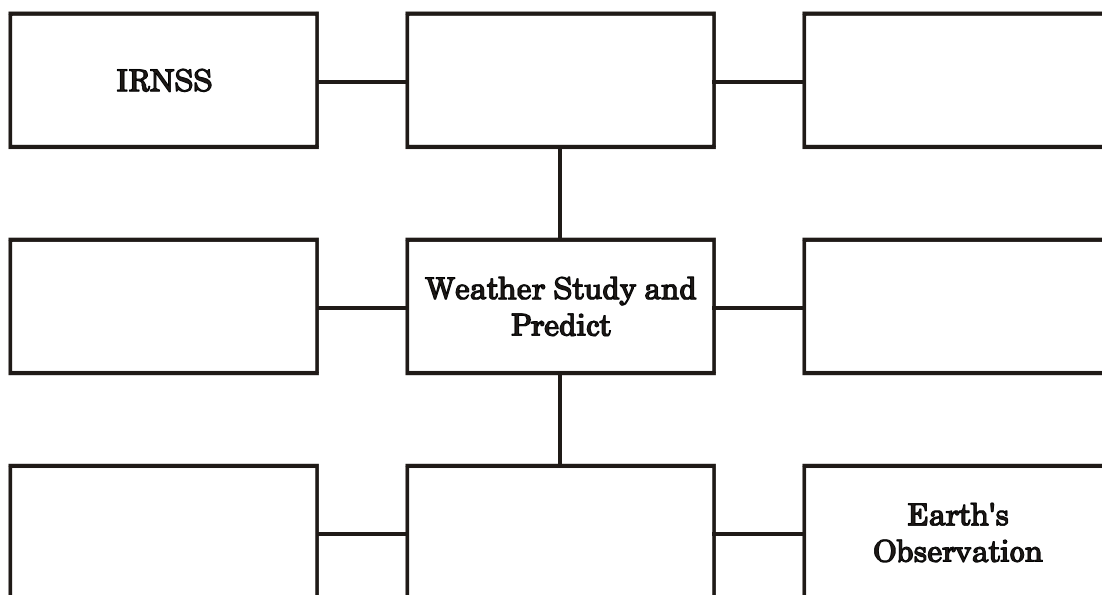
of water from 20°C to 100°C ?

[$c = 1 \text{ kcal/kg}^{\circ}\text{C}$]

(vi) Write the Cartesian sign convention for lenses.

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(vii) Complete the following chart :



(viii) Give the chemical equations for the following reactions :

- (a) Thermit reaction
- (b) Burning of magnesium in air
- (c) Iron nail/dust is dropped in aqueous solution of copper sulphate.

4. Answer the following (any *one*) :

5

- (i) "A rainbow is the combined effect of the refraction, dispersion and total internal reflection of light." Prove this statement with the help of diagram and explain it.

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(ii) Complete the following table :

Sr. No.	Straight chain of carbon atoms	Molecular formula	Name
1.	C	CH ₄	
2.	C-C		Ethane
3.	C-C-C	C ₃ H ₈	
4.	C-C-C-C		Butane
5.	C-C-C-C-C	C ₅ H ₁₂	
6.	C-C-C-C-C-C		Hexane
7.	C-C-C-C-C-C-C		Heptane
8.	C-C-C-C-C-C-C-C	C ₈ H ₁₈	
9.	C-C-C-C-C-C-C-C-C		Nonane
10.	C-C-C-C-C-C-C-C-C-C		Decane